

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.	BL1 – Remember	5

2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: - courses, including number, title, credits, syllabus, and prerequisites; - course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; - students, including student-id, name, and program; - instructors, including identification number, name, department, and title. - Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5
4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5

5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. ☐ A department controls a number of PROJECT. Projects have a unique name, number and a single location. ☐ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. ☐ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
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Code of Conduct:

*I **HARSHAN I / 1925260054** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Harshan

RUBRICS FOR CALCULATION :

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to Understand

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ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

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Activity Tool : Short Answer Text (High)

Assessment Tool Weightage: 30%

1)UPS Product Tracking Information System

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system

ANS)

The UPS Product Tracking System is designed to maintain accurate information about every shipped item and its current location. The main entity in the system is the Shipped Item, which contains details such as Item Number (Primary Key), Weight, Dimensions, Insurance Amount, Destination, and Final Delivery Date.

Each shipped item is received at one Retail Center. A Retail Center is identified by its Unique ID (Primary Key) and also stores its Type and Address. Since one retail center can receive many shipped items, the relationship between Retail Center and Shipped Item is One-to-Many (1:M).

After being received, each shipped item is transported through one or more Transportation Events, such as flights or truck deliveries. A Transportation Event is described by Schedule Number (Primary Key), Type, and Delivery Route. A shipped item can pass through multiple transportation events, and each transportation event can carry multiple shipped items. Therefore, the relationship between Shipped Item and Transportation Event is Many-to-Many (M:N).

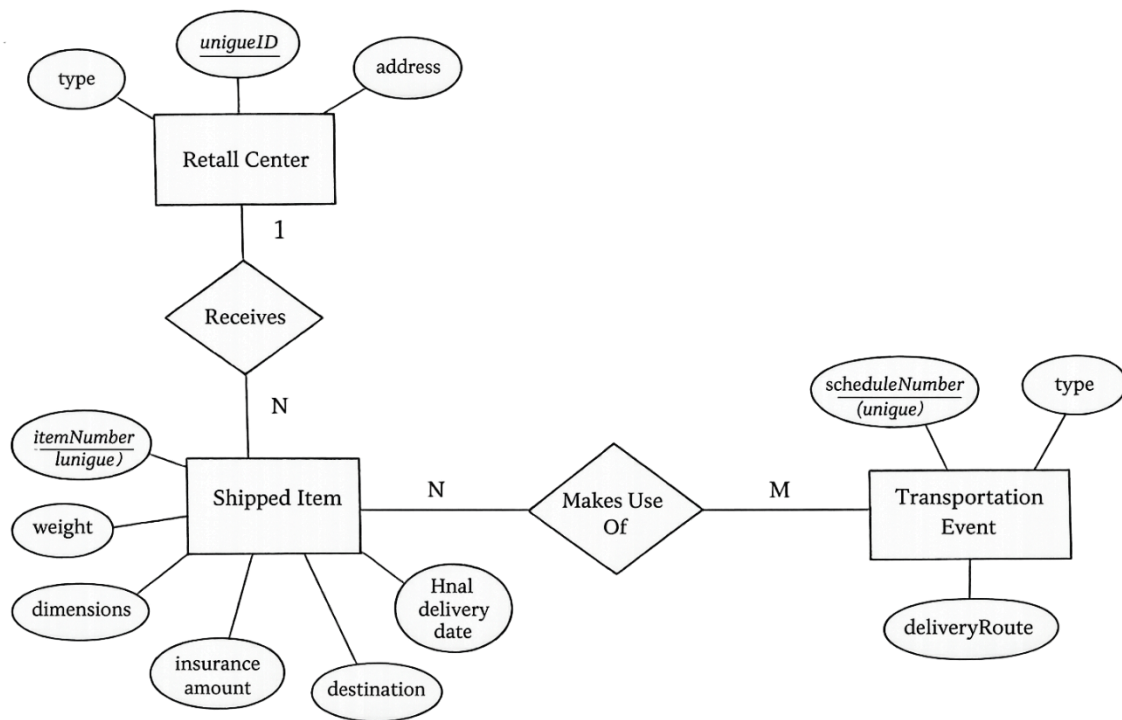
Thus, the ER diagram consists of three entities:

- Shipped Item
- Retail Center
- Transportation Event

The relationships are:

- Retail Center receives Shipped Item (1:M)
- Shipped Item is transported by Transportation Event (M:N)

This ER model efficiently represents the UPS product tracking system by storing shipment details, receipt information, and transportation history.



2) Course Management System

Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as

- course with attributes name, department, and c-number
- section with attributes s-number and enrollment, and dependent as a weak entity set on course
- room with attributes r-number, capacity, and building

Show an E-R diagram illustrating the use of all three additional entity sets listed.

ANS)

The E-R diagram for the **Course Management System** consists of four entities: **Course**, **Section**, **Exam**, and **Room**.

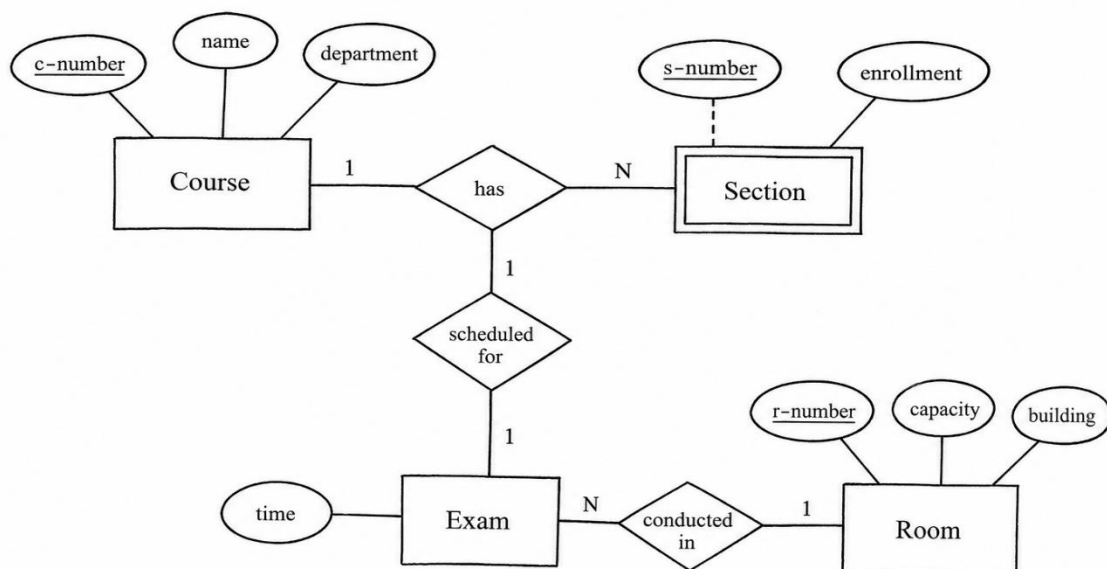
- **Course** is a strong entity with attributes **c-number (Primary Key)**, **name**, and **department**.
- **Section** is a **weak entity** that depends on the **Course** entity. It has the attributes **s-number (Partial Key)** and **enrollment**.

- **Exam** is an entity with the attribute **time**. Each section is associated with one final exam.
- **Room** is a strong entity with the attributes **r-number (Primary Key)**, **capacity**, and **building**.

The relationships are:

- A **Course** can have **many Sections (1:N)**.
- Each **Section** is **scheduled for one Exam (1:1)**.
- Each **Exam** is **conducted in one Room**, while a **Room** can be used for multiple exams at different times (N:1).

Thus, the E-R diagram effectively represents the scheduling of final examinations by linking courses, sections, exams, and rooms while maintaining data integrity through appropriate entities and relationships.



3) University Registrar's Office Management System

A university registrar's office maintains data about the following entities:

- (f) courses, including number, title, credits, syllabus, and prerequisites;
- (g) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;
- (h) students, including student-id, name, and program;
- (i) instructors, including identification number, name, department, and title.
- (j) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.

Construct an E-R diagram for the registrar's office.

ANS)

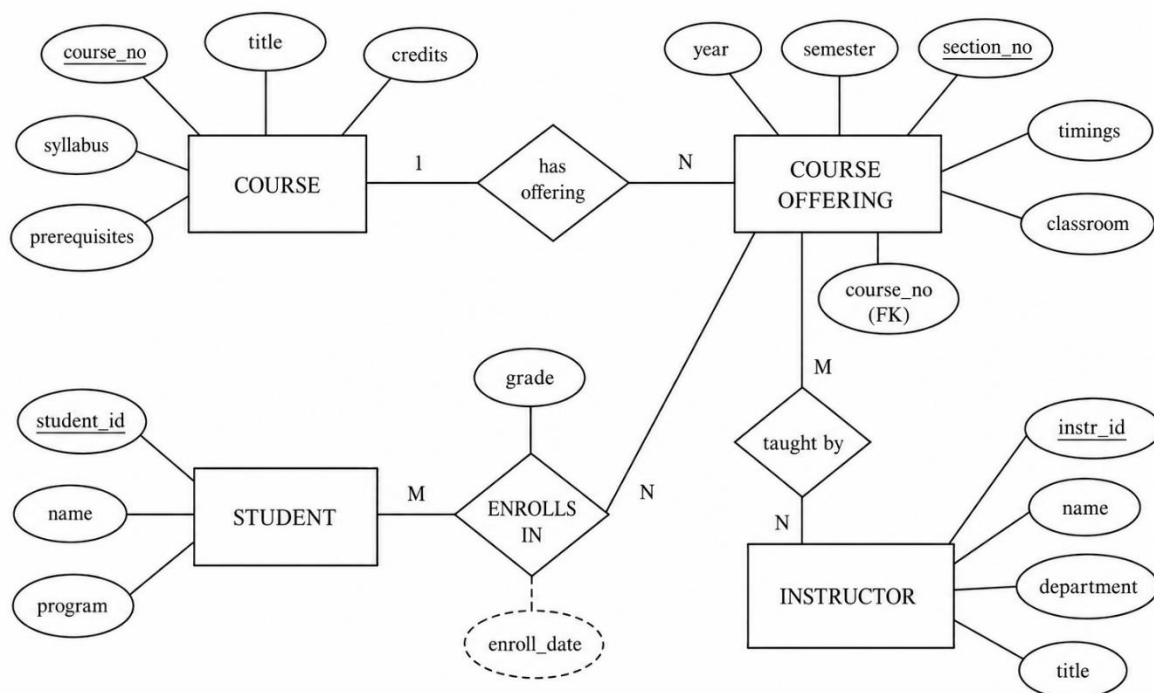
The E-R diagram for the University Registrar's Office Management System consists of four main entities: Course, Course Offering, Student, and Instructor.

- Course is a strong entity with the attributes course number (Primary Key), title, credits, syllabus, and prerequisites.
- Course Offering represents a specific offering of a course and includes the attributes course number, year, semester, section number, timings, and classroom. Each course can have multiple course offerings.
- Student is a strong entity with the attributes student-id (Primary Key), name, and program.
- Instructor is a strong entity with the attributes identification number (Primary Key), name, department, and title.

The relationships are:

- A Course can have many Course Offerings (1:N).
- An Instructor can teach one or more Course Offerings, and a Course Offering may be taught by one or more Instructors (M:N).
- A Student can enroll in many Course Offerings, and each Course Offering can have many Students (M:N). The Enrollment relationship stores the grade awarded to each student for the corresponding course offering.

Thus, the E-R diagram models the university registrar's office by representing courses, course offerings, students, instructors, enrollments, and grades, along with their relationships.



4)Hospital Management System

Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors.

Associate with each patient a log of the various tests and examinations conducted.

Construct appropriate tables for the above ER Diagram :

Patient(SS#, name, insurance)

Physician (name, specialization)

Test-log(SS#, test-name, date, time)

Doctor-patient (physician-name, SS#)

Patient-history(SS#, test-name, date)

ANS)

The E-R diagram for the Hospital Management System consists of four main entities: Patient, Physician, Test-Log, and Patient-History.

- Patient is a strong entity with the attributes SS# (Primary Key), name, and insurance.
- Physician is a strong entity with the attributes name (Primary Key) and specialization.
- Test-Log records all medical tests conducted for patients and contains the attributes SS#, test-name, date, and time.
- Patient-History stores the history of tests undergone by patients and includes the attributes SS#, test-name, and date.

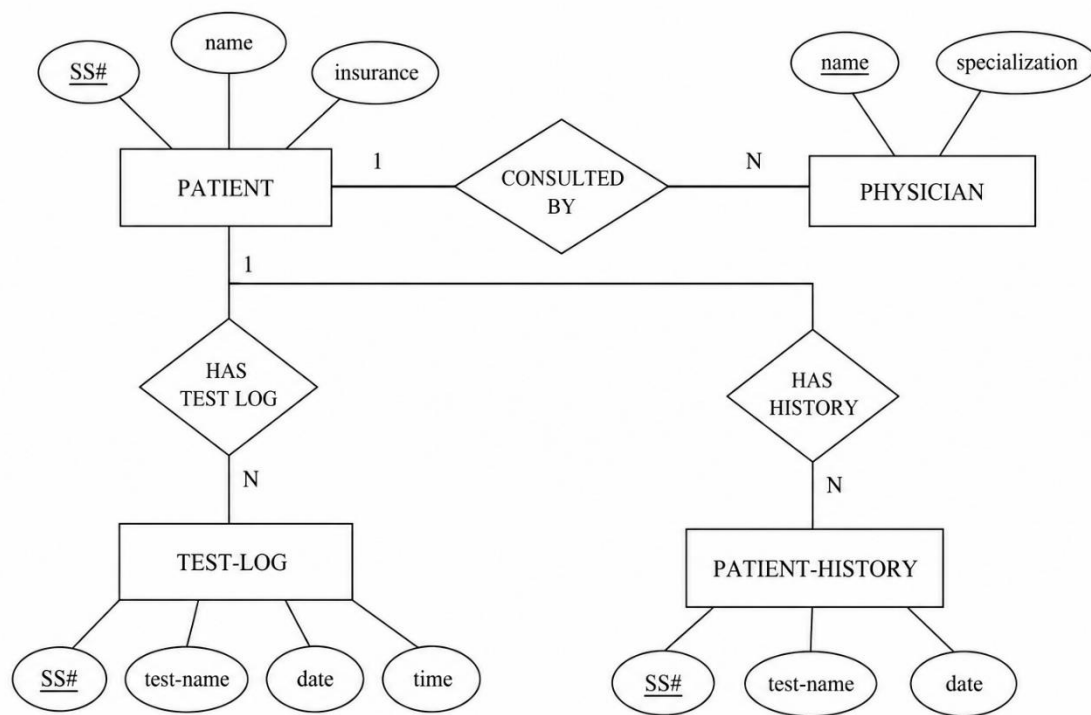
Relationships

- A Physician can treat many Patients, and a Patient can be treated by many Physicians through the Doctor-Patient relationship (M:N).
- A Patient can have multiple Test-Log records, while each Test-Log belongs to one Patient (1:N).
- A Patient can have multiple Patient-History records, while each Patient-History record belongs to one Patient (1:N).

Relational Tables

- Patient (SS#, name, insurance)
- Physician (name, specialization)
- Test-Log (SS#, test-name, date, time)
- Doctor-Patient (physician-name, SS#)
- Patient-History (SS#, test-name, date)

Thus, the E-R diagram represents the hospital database by maintaining information about patients, physicians, medical tests, patient history, and the doctor–patient relationship efficiently.



5) Company Project Management System

Construct an ER Diagram for Company having following details :

Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations.

□ A department controls a number of PROJECT. Projects have a unique name, number and a single location.

□ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee.

□ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).

ANS)

The E-R diagram for the **Company Project Management System** consists of four main entities: **Department**, **Project**, **Employee**, and **Dependent**.

- **Department** is a strong entity with the attributes **department-name (Primary Key)** and **locations**. Each department is managed by one employee, and the **manager's start date** is recorded. A department may have **multiple locations**.
- **Project** is a strong entity with the attributes **project-number (Primary Key)**, **project-name**, and **location**. Each project is controlled by one department.
- **Employee** is a strong entity with the attributes **SSN (Primary Key)**, **name**, **address**, **salary**, **sex**, and **birth date**. Every employee belongs to one department but may work on multiple projects. The **number of hours worked per week** on each project is stored. Each employee also has an **immediate supervisor** (recursive relationship).
- **Dependent** is a **weak entity** that depends on the Employee entity. It has the attributes **dependent-name (Partial Key)**, **birth date**, and **relationship**.

Relationships

- A **Department** employs **many Employees**, while each **Employee** belongs to **one Department (1:N)**.
- A **Department** controls **many Projects**, while each **Project** is controlled by **one Department (1:N)**.
- An **Employee** manages **one Department**, and each **Department** has **one Manager**. The relationship stores the **start date**.
- An **Employee** can work on **many Projects**, and a **Project** can have **many Employees (M:N)**. The relationship includes the attribute **hours/week**.
- An **Employee** supervises **many Employees**, while each employee has **one immediate supervisor (1:N recursive relationship)**.
- An **Employee** can have **many Dependents**, while each **Dependent** belongs to **one Employee (1:N)**.

